

MATH230 HW2

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Exercise. 3.3.

Let W be a random variable giving the number of heads minus the number of tails in three tosses of a coin. List the elements of the sample space S for the three tosses of the coin and to each sample point assign a value w of W .

X

$W = \#H - \#T$ in three tosses of a coin.

$$\begin{array}{c}
 H < H < H \\
 H < H < T \\
 H < T < H \\
 H < T < T \\
 T < H < H \\
 T < H < T \\
 T < T < H \\
 T < T < T
 \end{array}$$

$$\Rightarrow S = \{ \overset{3}{HHH}, \overset{1}{HHT}, \overset{1}{HTH}, \overset{1}{HTT}, \overset{1}{THH}, \overset{-1}{THT}, \overset{-1}{TTH}, \overset{-3}{TTT} \}$$

Exercise. 3.8.

Find the probability distribution of the random variable W in **Exercise 3.3**, assuming that the coin is biased so that a head is twice as likely to occur as a tail.

PM: $\begin{matrix} S' & P \\ T & \rightarrow \alpha \\ H & \rightarrow 2\alpha \end{matrix}$

of each toss

$HHH \rightarrow \frac{8}{27}$
 $HHT \rightarrow \frac{4}{27}$
 $HTT \rightarrow \frac{2}{27}$
 THT
 TTT

$HTH \rightarrow \frac{4}{27}$
 $TTH \rightarrow \frac{4}{27}$
 $TTT \rightarrow \frac{1}{27}$

So $P(W=3) = P(W=1/34)$
 $= P(\cancel{HHH}) = \frac{8}{27}$

$P(W=1) = \frac{12}{27}$
 $P(W=-1) = \frac{6}{27}$
 $P(W=-3) = \frac{1}{27}$

1

Since $P(S) = P(\{T, H\}) = 1$
 & each event is disjoint
 $\Rightarrow S' = \{T\} \cup \{H\}$
 $\Rightarrow P(S) = P(\{T\} \cup \{H\})$
 $= P(\{T\}) + P(\{H\})$
 $= 3\alpha = 1. \quad \alpha = \frac{1}{3} \text{ gives}$

$P(\{T\}) = \frac{1}{3}$
 $P(\{H\}) = \frac{2}{3}$

Exercise. 3.23.

Find the cumulative distribution function of the random variable W in **Exercise 3.8**. Using $F(w)$, find

(a) $P(W > 0)$; $\swarrow$

(b) $P(-1 \leq W < 3)$. $\searrow$

$$P(W > 0) = P(W=1) + P(W=3) = \frac{20}{27}$$

$$\hookrightarrow P(W=-1) + P(W=1) = \frac{16}{27} = \left(\frac{2}{3}\right)$$

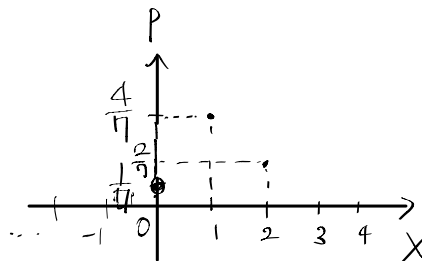
Exercise. 3.11.

2+3

A shipment of 7 televisions contains 2 defective ones. A hotel makes a random purchase of 3 televisions. If x is the number of defective ones purchased by the hotel, find the probability distribution of X . Express the results graphically as a probability histogram.

$$nC_3 = 35$$

$$\begin{aligned} P(X=3) &= 0 \text{ since } P(X \in \{34\}) = P(\emptyset) \\ P(X=2) &= \frac{2C_2 \cdot 5C_1}{nC_3} = \frac{1}{7} \\ P(X=1) &= \frac{2C_1 \cdot 5C_2}{nC_3} = \frac{4}{7} \\ P(X=0) &= \frac{5C_3}{nC_3} = \frac{2}{7} \end{aligned}$$



Exercise. 3.15

Find the cumulative distribution function of the random variable X representing the number of defectives in **Exercise 3.11**. Then using $F(x)$, find

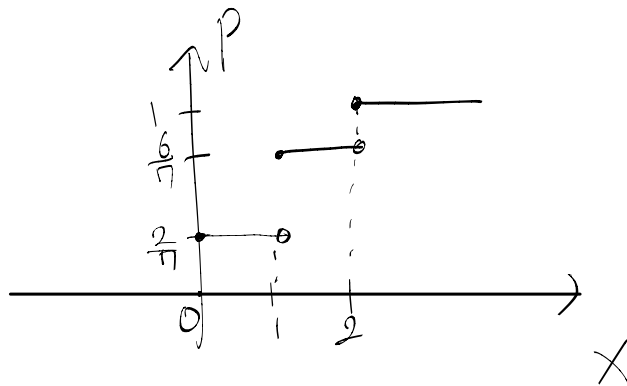
(a) $P(X=1)$; $\rightarrow \frac{4}{7}$

(b) $P(0 < X \leq 2)$.

$$\hookrightarrow \frac{3}{7}$$

Exercise. 3.16.

Construct a graph of the cumulative distribution function of **Exercise 3.15**.



Exercise. 3.14.

The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with a cumulative distribution function

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-4x} & x \geq 0 \end{cases} \rightarrow \frac{1}{5}$$

Find the probability of waiting fewer than 12 minutes between successive speeders

(a) using the cumulative distribution function of X . $e^{-4x} \stackrel{?}{\geq}$

(b) using the probability density function of X .

$$F(12) = 1 - e^{-0.8} = 0.55$$

↳ Since $F(x)$ is continuous, $F' = f = \begin{cases} 0 & x < 0 \\ 4e^{-4x} & x \geq 0 \end{cases}$

$$\begin{aligned} P(X \leq 12) &= \int_{-\infty}^{0.2} f = \int_{-\infty}^0 f + \int_0^{0.2} 4e^{-4x} dx \\ &= \left[-e^{-4x} \right]_0^{0.2} = \boxed{1 - e^{-0.8}} \end{aligned}$$

Exercise. 3.29.

An important factor in solid missile fuel is the particle size distribution. Significant problems occur if the particle sizes are too large. From production data in the past, it has been determined that the particle size *in micrometers* distribution characterized by

$$f(x) = \begin{cases} 3x^{-4}, & x > 1, \\ 0, & \text{elsewhere.} \end{cases}$$

outways $f(x) \geq 0$ o it's integrate

(a) Verify that this is a valid probability function.

$$\int_{-\infty}^{\infty} f(x) dx = \int_1^{\infty} 3x^{-4} dx \Rightarrow \text{valid.}$$

(b) Evaluate $F(x)$.

$$= \lim_{x \rightarrow \infty} \left\{ \int_1^x 3(x')^{-4} dx' \right\}$$

(c) What is the probability that a random particle from the manufactured fuel exceeds 4 micrometers?

$$= \boxed{1}$$

$\Rightarrow$ OK.

$\downarrow$
 $P(X > 4)$?

$$F(x) = \int_{-\infty}^x f(x') dx'. \text{ So } \forall x \leq 1,$$

$$\int_{-\infty}^x f(x') dx' = 0 = F(x).$$

$$\forall x > 1,$$

$$\begin{aligned} \int_1^x f(x') dx' &= \int_1^x 3(x')^{-4} dx' \\ &= \left[-\frac{1}{3} (x')^{-3} \right]_1^x = -\frac{1}{3x^3} + 1. \end{aligned}$$

$$\text{So } F(x) = \begin{cases} -\frac{1}{3x^3} + 1, & x > 1 \\ 0, & x \leq 1. \end{cases}$$

$$= 1 - F(4)$$

$$= \left. 1 + \frac{1}{x^3} \right|_{x=4} - 1 = \frac{1}{64}$$